

EDUGENIE

5.3 Automated Assessment & Quest Generation

EDUGENIE can transform learning objectives into interactive tasks or quests. For example, a chemistry objective could become a task requiring students to identify elements and combine them correctly.

Assessment does not have to depend entirely on traditional written examinations. The system could record decisions, attempts, errors, completion time, and problem-solving strategies.

Teachers could use these records to understand how students approached a task. This creates an opportunity for continuous assessment rather than assessment only at the end of a lesson.

Key Points

- Generates tasks from learning objectives.
- Records decisions and attempts.
- Tracks errors and completion time.
- Captures problem-solving strategies.
- Supports continuous assessment.

Conceptual Picture / Workflow

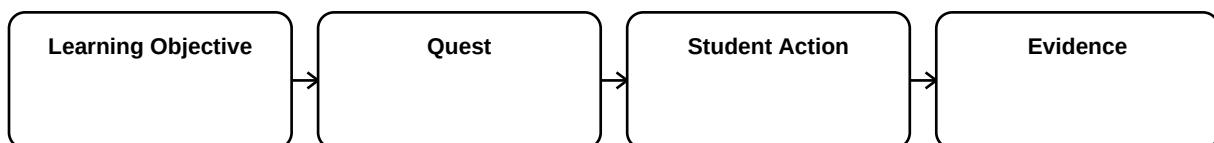


Figure: Simplified representation of the learning mechanism.

Comparison / Design Table

Objective	Traditional assessment	Interactive quest
Chemistry	Written questions	Identify and combine elements
Physics	Numerical test	Solve an interactive motion challenge
Biology	Label a diagram	Explore an organ system and answer in-world tasks
Evidence	Final answer	Actions, attempts, decisions and outcome

Sample Coding Concept

```
# Conceptual quest generation and assessment
```

```
objective = "Understand covalent bonding"

quest = generate_quest(objective)

student_result = run_quest(quest)

record = {
    "decisions": student_result.decisions,
    "attempts": student_result.attempts,
    "errors": student_result.errors,
    "time": student_result.completion_time
}

teacher_dashboard.save(record)
```

Example Automated Quest

For a chemistry lesson, the teacher can provide the objective “understand covalent bonding.” EDUGENIE can generate an interactive task where learners identify suitable atoms, explore bonding relationships, complete the activity, and receive feedback. The teacher can review the recorded learning evidence.